

GEO spacecraft charging

GEO: not in the ionosphere anymore

- ❖ Which space environment is relevant?

$$T_e \sim 1000 \text{ eV} \sim T_e \sim 10^7 \text{ K}$$

$$V_{sc} \sim \sqrt{\frac{GM}{r}} \sim 3 \text{ km/s}$$

$$V_{e+th} = 10,000 \text{ km/s}$$

$$V_{i+th} \sim 460 \text{ km/s} \quad (p^T, T_p > 4T_e)$$

$$T_{ph} = 0$$

$$I_e = -\frac{1}{4} q_e n_e v_{e,th} A_e \left(e^{q_e V / k_B T_e} \right)$$

Eq (21)

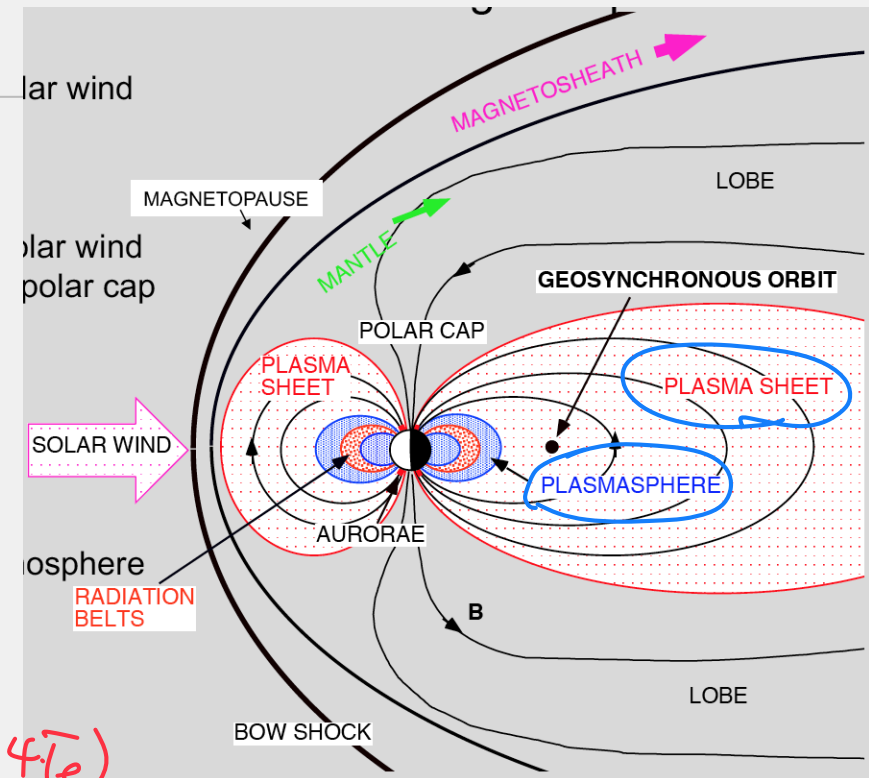
$$I_i = \frac{1}{4} q_e n_i v_{i,th} A_i \left(1 - \frac{q_e V}{k_B T_i} \right)$$

Eq (25)

$I_e + I_i$

$$V_{fl} = \frac{k_B T_e}{q_e} \ln \left[\sqrt{\frac{m_e}{m_i}} \left(1 - \frac{q_e V}{k_B T_i} \right) \right]$$

Eq (26)



$$V_A = -2500 \text{ V}$$

GEO spacecraft charging

in sunlight side

e^- $I_e = -\frac{1}{4}q_e n_e v_{e,th} A_e \left(1 + \frac{q_e V}{k_B T_e}\right)$ for $V > 0$

p^+ $I_i = \frac{1}{4}q_e n_0 v_{i,th} A_i \left(e^{-q_e V/k_B T_i}\right)$ for $V > 0$

$I_{ph} = ?$

$I_e + I_i + I_{ph} = 0$

What about the photocurrent?

- ❖ Photocurrent is electrons leaving the spacecraft surface due to photoelectric effect from solar UV
- ❖ Electrons going **away from** spacecraft = current **towards** spacecraft

$$I_{ph} = J_0 A_n \cos \theta e^{-qV / kT_{ph}}$$

$$kT_{ph} \sim 2 \text{ eV}$$

Eq (28)

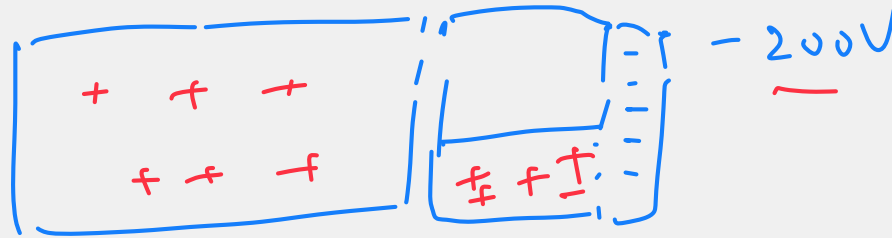
$$I_e + I_i + I_{ph} \Rightarrow V_{fl}$$

← sunlight

Differential Charging

making it worse

material dependent



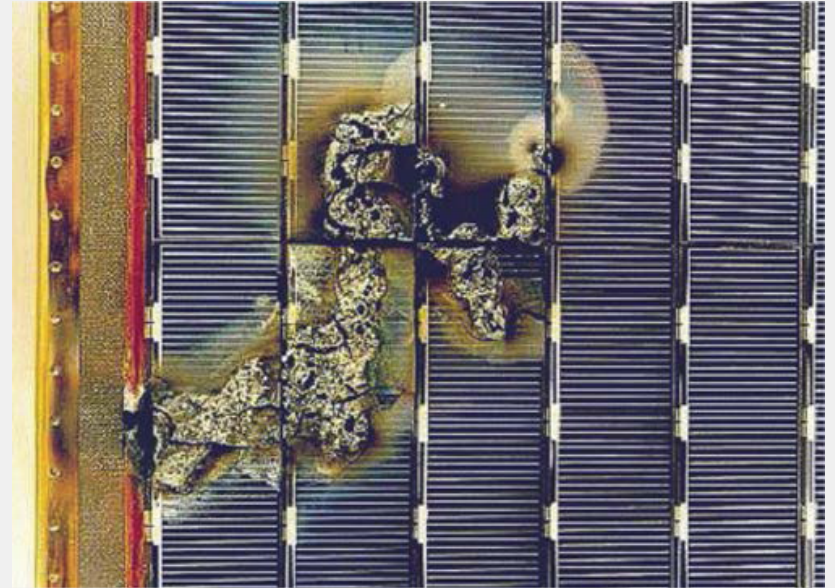
+ 5 Volt

sun light

Three red arrows originate from the bottom left and point upwards towards the underlined text 'sun light'.

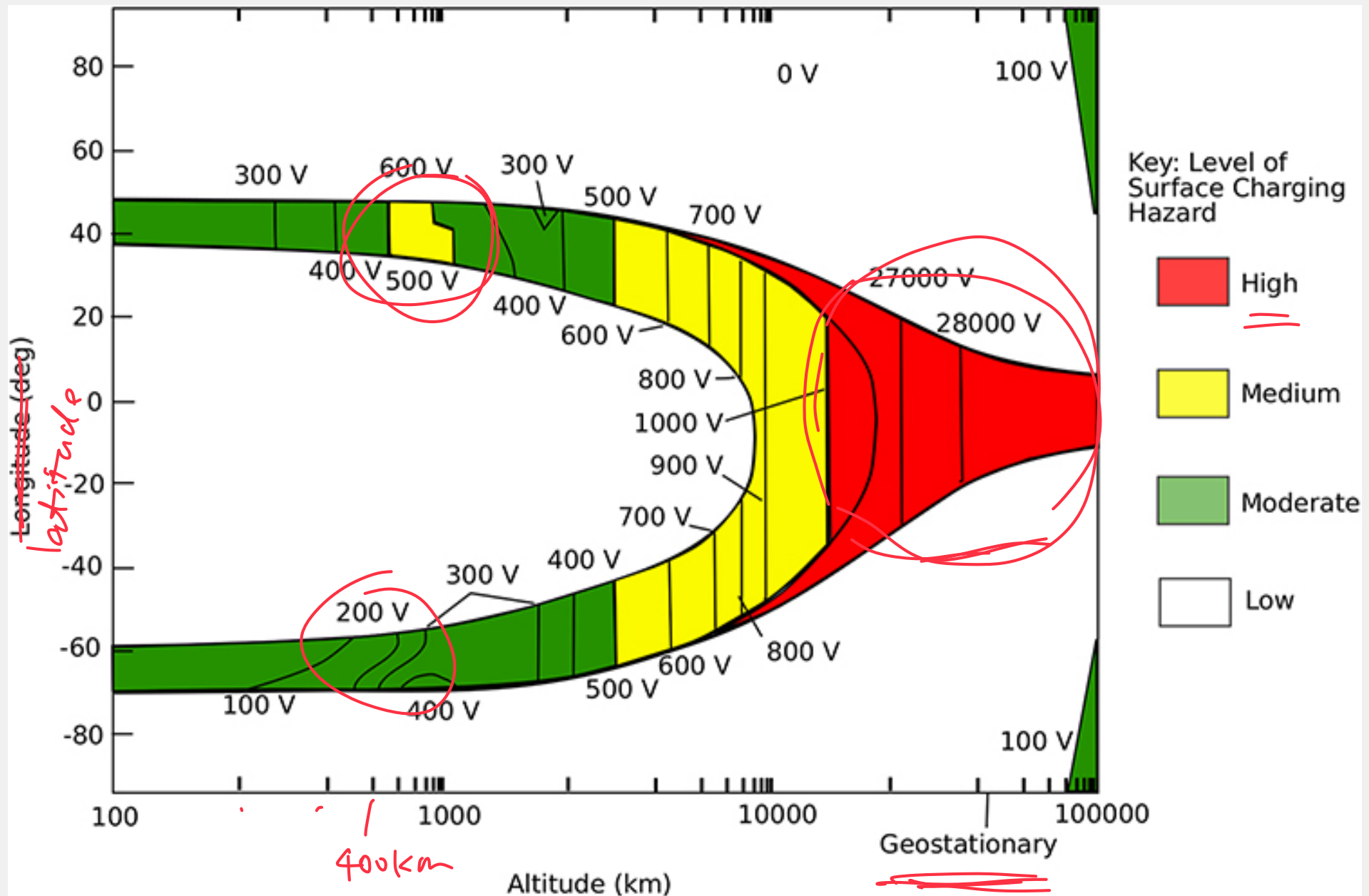
Effects of Spacecraft Charging

- ❖ We rarely care about the absolute potential: simply relative to "space"
- ❖ Effects occur when there is significant potential difference (i.e. voltage) between two regions of spacecraft
- ❖ **Electrostatic Discharge:** Transfer of charge through spacecraft material due to electric field
 - ❖ Starts at -150 to -500 V
 - ❖ Produces electromagnetic pulse / interference
- ❖ **Dielectric Breakdown:** materials break down at some electric field (air: 3 MV/m at sea level)



← 100s of keV up to
MeV e^-

Where is spacecraft charging a concern?



LEO Discharge events occur over South America

❖ And at high Latitudes (~70 degrees).

❖ Why?

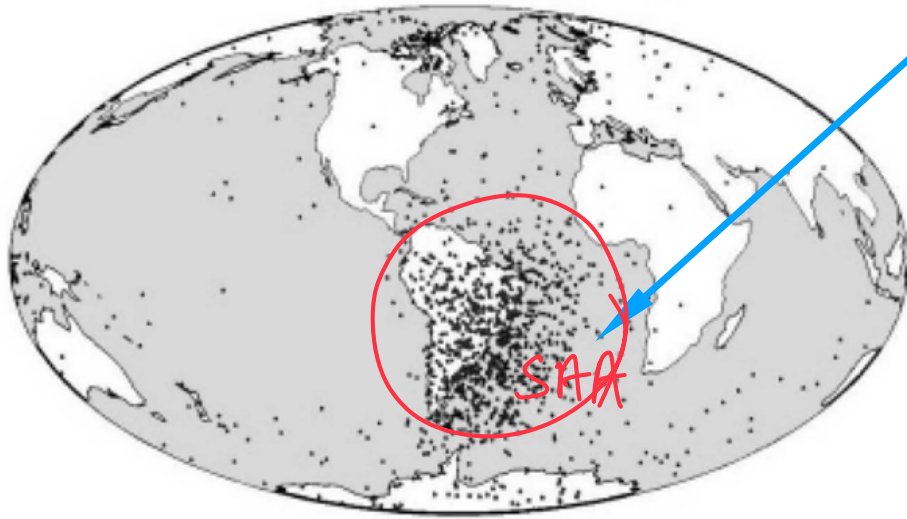


Figure 5. Single-event upsets on the low-altitude UoSAT-2 amateur-radio communications satellite. The majority, from protons in the inner radiation belt, occur in the South Atlantic Anomaly over South America and the South Atlantic Ocean. Upsets scattered over the rest of the globe are from galactic cosmic rays [6] (courtesy of NOAA).

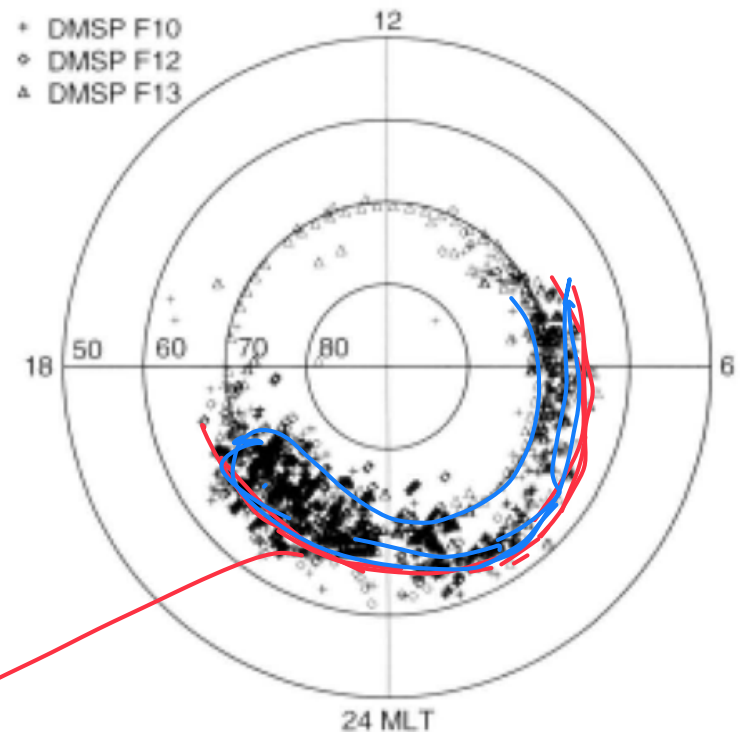
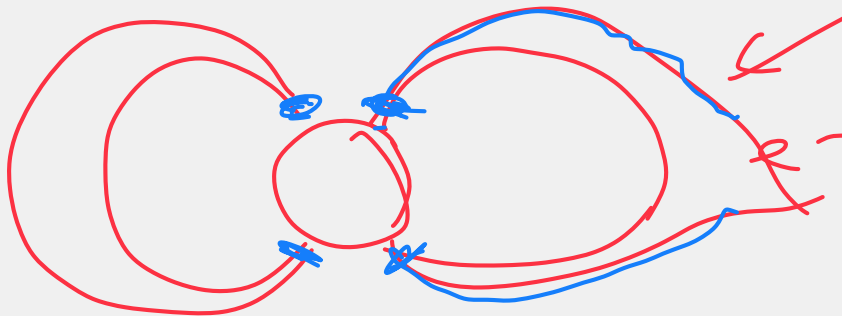


Figure 8. The location in magnetic latitude and magnetic local time of surface-charging events measured on the DMSP satellites at low altitude in the auroral zones [22] (courtesy of the Air Force Research Laboratory).



GEO: Discharge events occur in the post-midnight and early morning...

❖ Why?

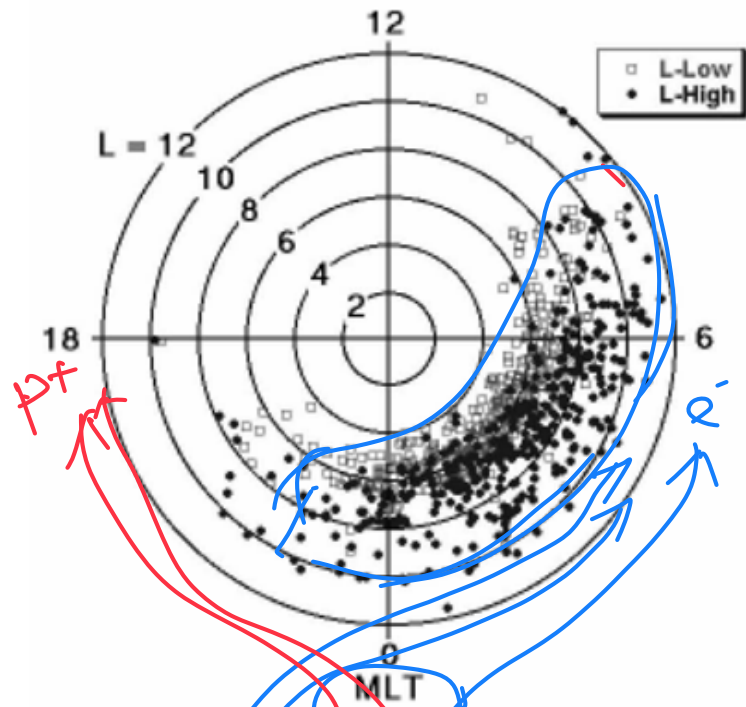
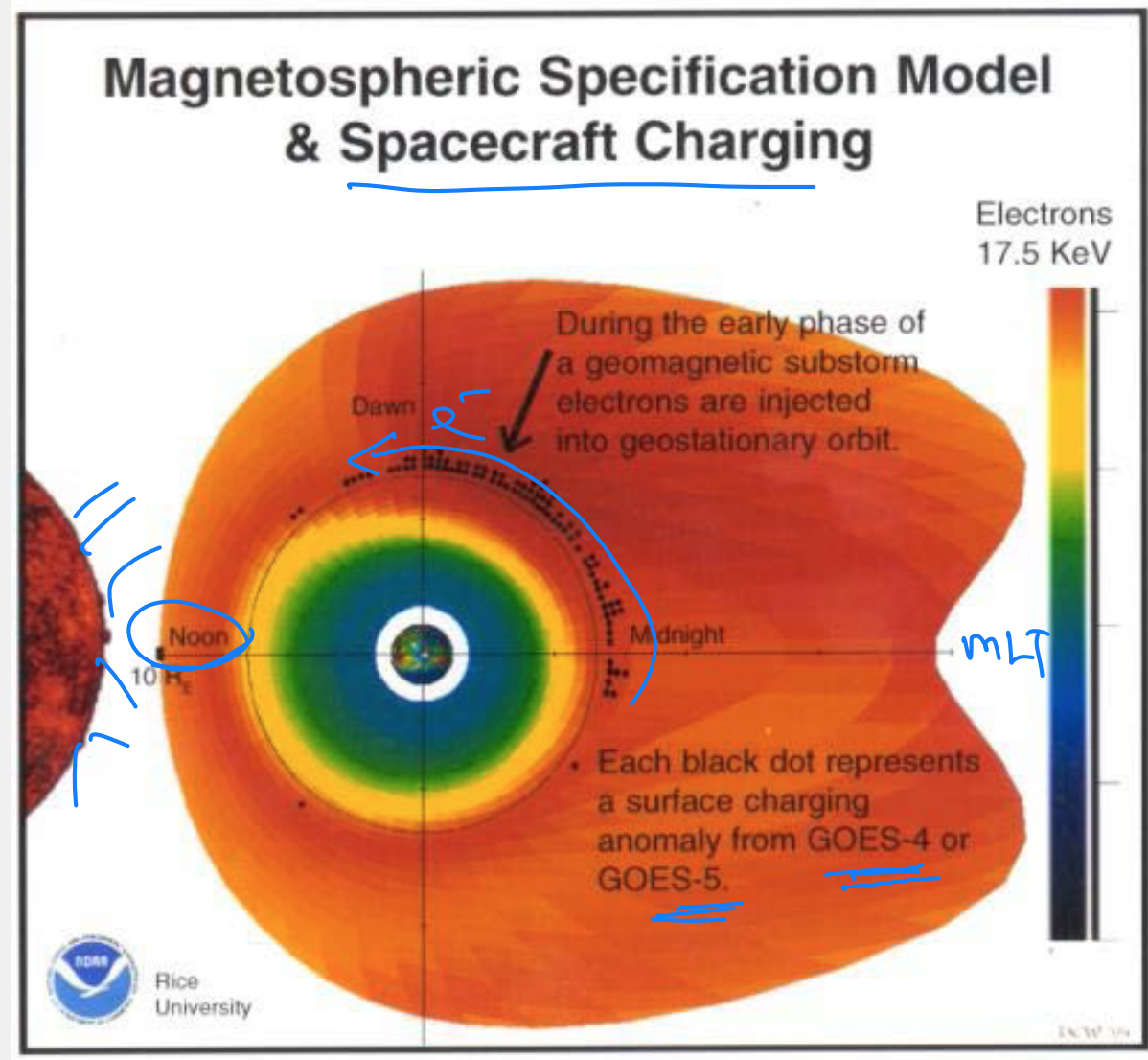


Figure 10. The location in Magnetic Local Time and L-shell where the spacecraft frame potential was less than -30 V in a Molniya orbit. The squares indicate the low L-shell limit of the charging (no data were taken below $L \sim 4$) and the circles show the high L-shell limit for each orbit. The points were not connected to simplify the drawing.



Radio Wave Propagation

- ❖ Time for some plasma physics!
 - ❖ Plasma oscillations
 - ❖ Plasma frequency
 - ❖ index of refraction (from Maxwell's equations)
 - ❖ Add collisions: absorption
 - ❖ MUF, LUF, enhanced EUV and X-ray effect

Plasma

oscillations

$$\sigma = -q_e n_e x$$

$$E = - \frac{q_e n_e x}{\epsilon_0} \quad \left(E = \frac{\sigma}{\epsilon_0} \right)$$

$$m_e a = F = q_e E$$

$$m_e \frac{dv}{dt} = m_e \frac{d^2 x}{dt^2} = - \frac{q_e^2 n_e}{\epsilon_0} x$$

$$\frac{d^2 x}{dt^2} + \left(\frac{q_e^2 n_e}{m_e \epsilon_0} \right) x = 0 \Rightarrow \omega_p^2$$

$$x(t) = A \cos(\omega_p t)$$

$$\omega_p = \sqrt{\frac{q_e^2 n_e}{m_e \epsilon_0}}$$

$$\propto \sqrt{n_e}$$

EM wave
 $v_p = \frac{c}{n}$

plasma, $n^2 = 1 - \frac{\omega_p^2}{\omega^2}$

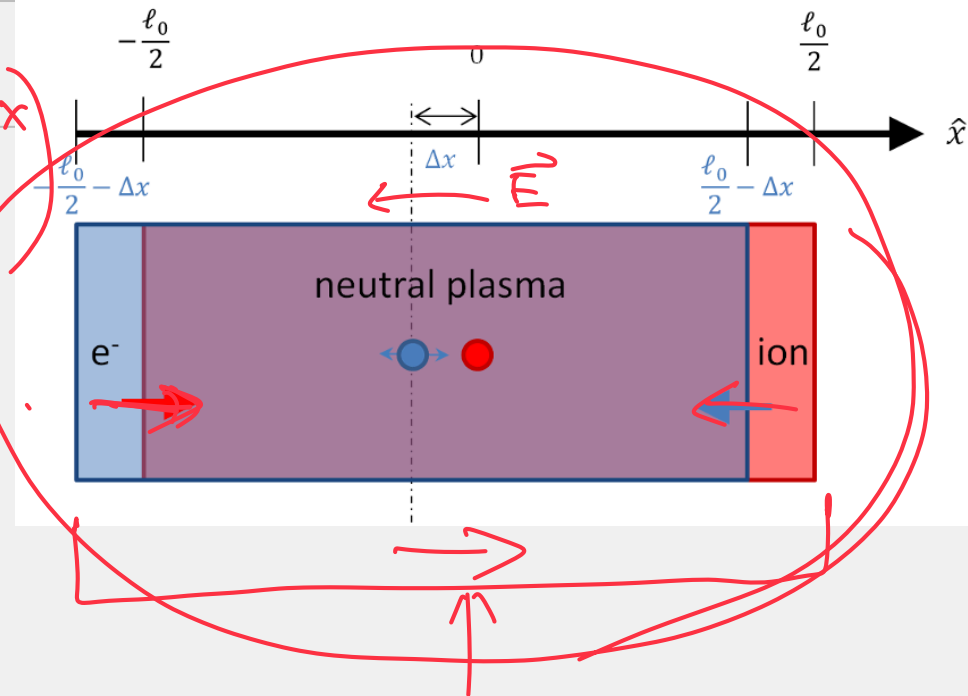
$$\omega = 2\pi f$$

radio wave freq

at $\omega = \omega_p$, $n = 0$
 $v_p \rightarrow \infty$

if $\omega < \omega_p$
 $n = \text{complex}$
 $n = \alpha + i\beta$

$\Rightarrow$ reflect

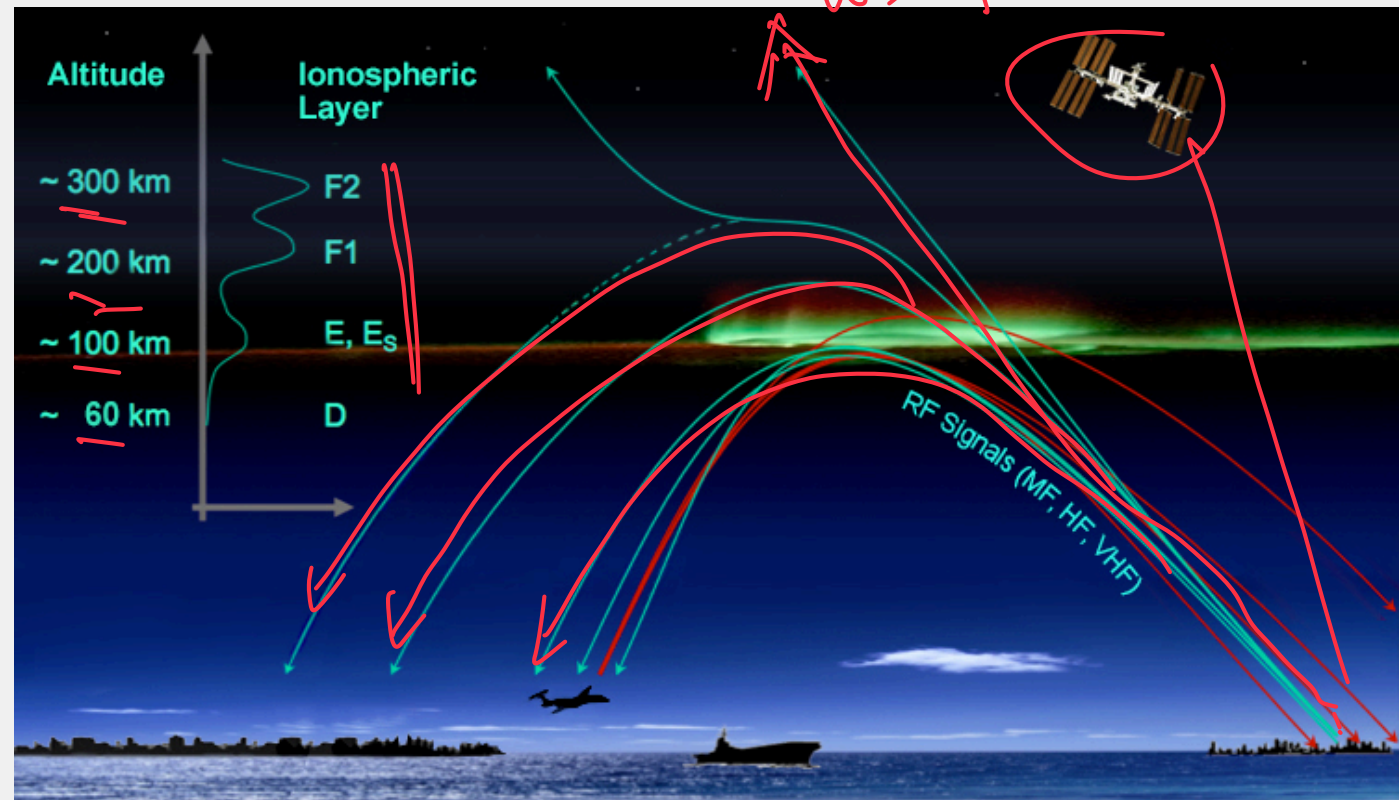


Reflection of EM Waves

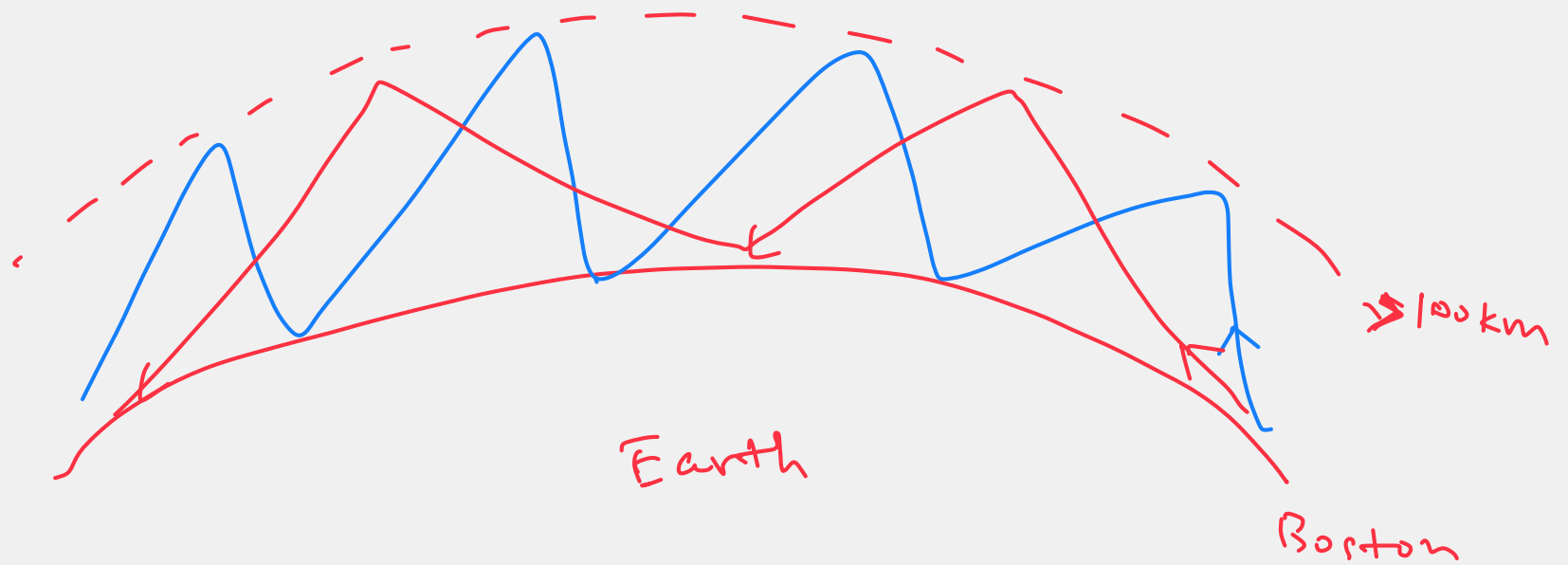
- ❖ Plasma frequency ω_p directly related to electron density $\omega_p \propto \sqrt{n_e}$
- ❖ Radio waves above ω_p pass through the ionosphere (with some loss); electrons cannot respond fast enough (but still have some oscillations)
- ❖ Radio waves below ω_p are reflected; electrons are "shaken" and re-radiate

- ❖ **Implications:**

- ❖ must use frequencies above ω_p to communicate to satellites
- ❖ Can communicate over-the-horizon with frequency near / below ω_p

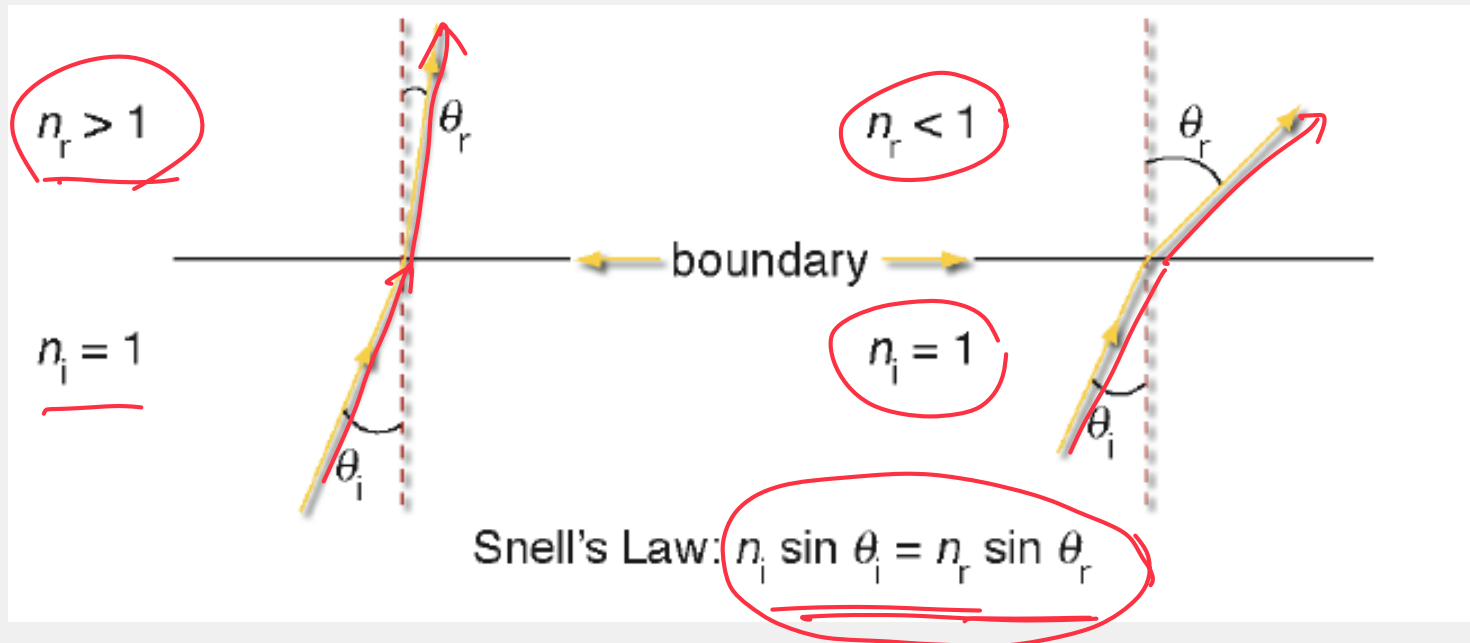


Over-the-horizon radar or communication



Index of Refraction

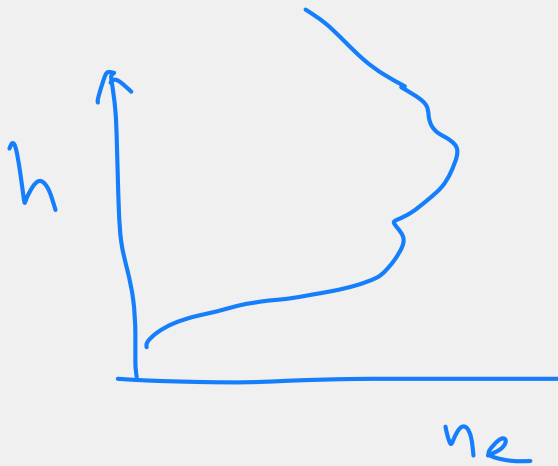
water: $n \sim 1.33$



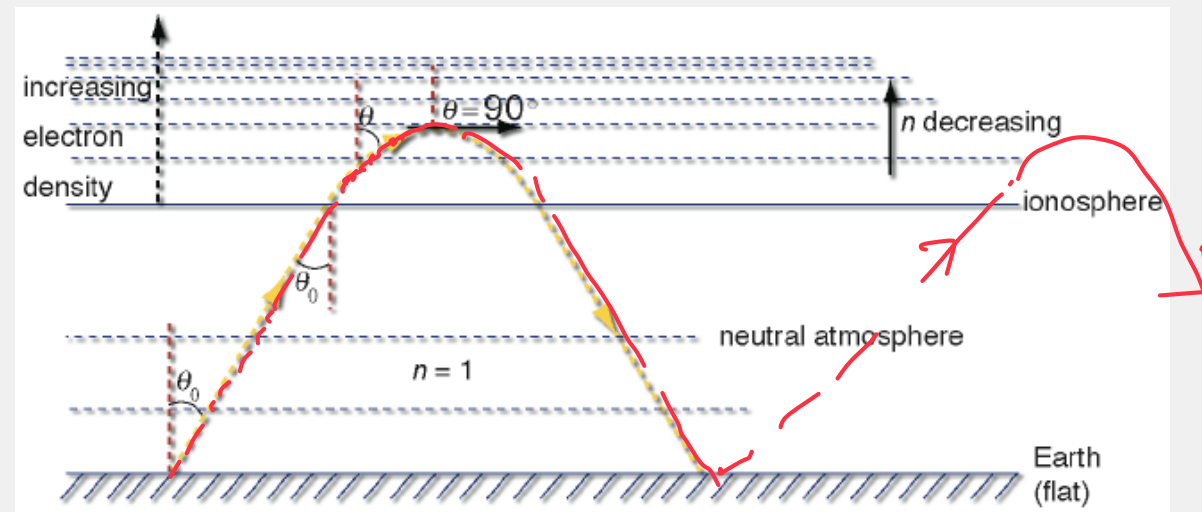
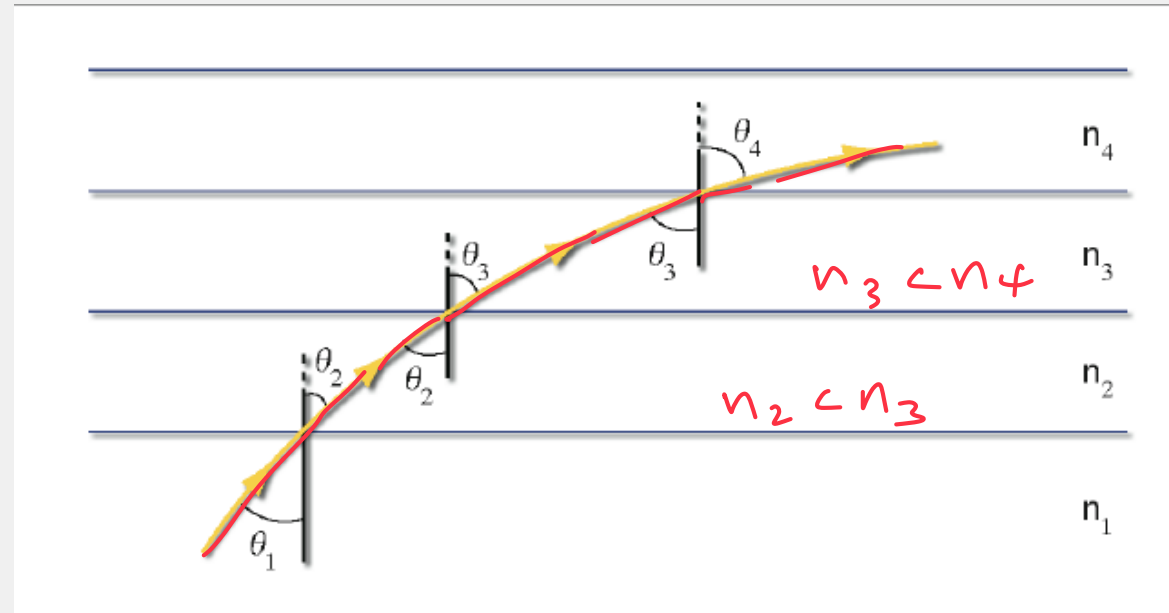
- ❖ As a radio wave propagates from one medium to another, its propagation direction is **refracted**, based on the **indices of refraction**, n_1 and n_2 .

Ionospheric Refraction

$$\underline{n^2 = 1 - \frac{\omega_p^2}{\omega^2}}$$

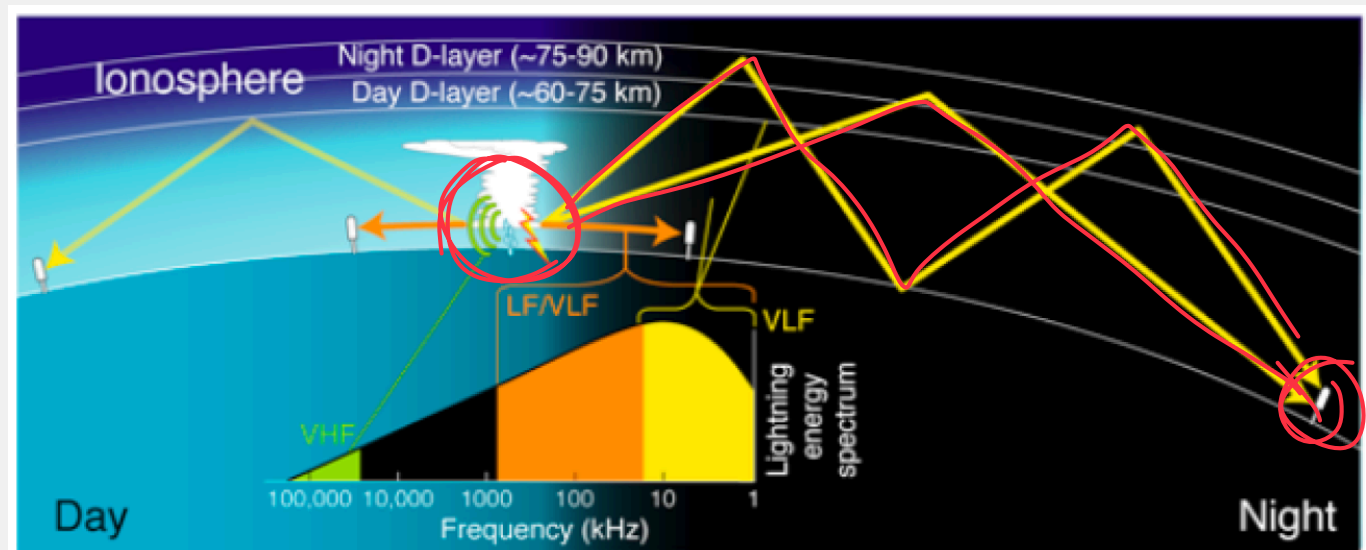


- ❖ We can treat the ionosphere as successive layers, and look at refraction from one layer to the next
- ❖ Continuous refraction
- ❖ End result: ray "bends" and turns back to the ground. Not a hard reflection!

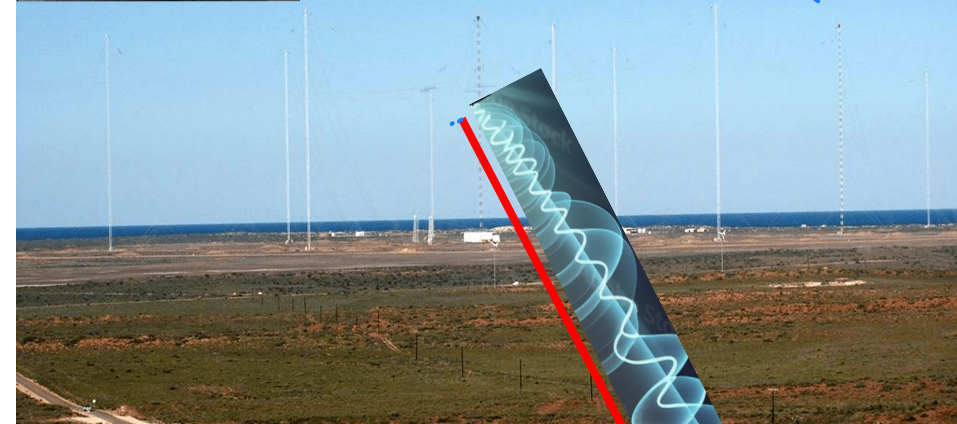
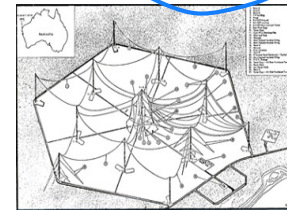
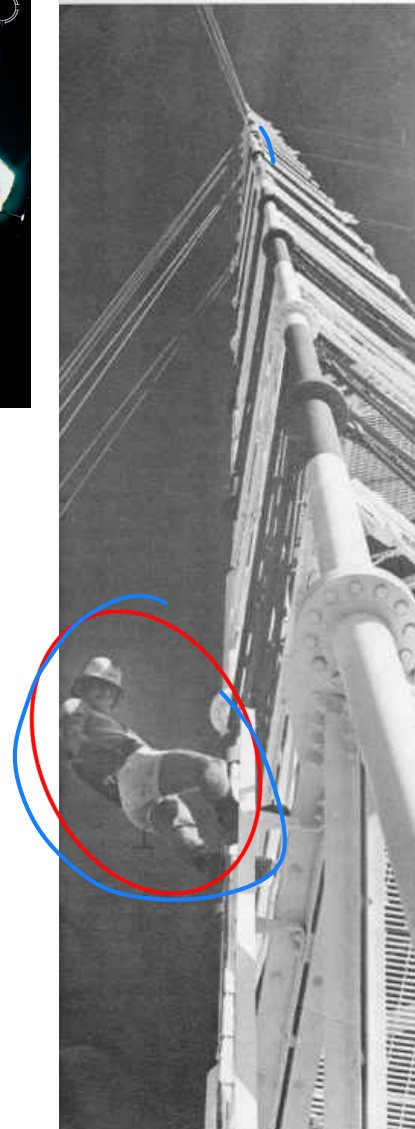
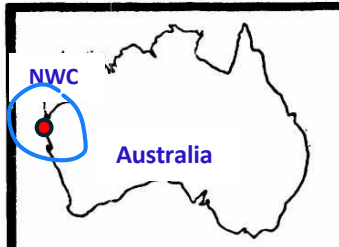


Critical Frequencies in the Ionosphere

- ❖ In the F-region, $f_c \sim 3-30$ MHz *f_c critical*
 - ❖ higher frequencies pass through the ionosphere, with some refraction
 - ❖ over-the-horizon radio
- ❖ In the E-region, $f_c \sim 1-2$ MHz *$\omega_p \propto \sqrt{n_e}$*
 - ❖ but sporadic-E increases f_c up to 100 MHz
- ❖ In the D-region, our simple model breaks. *$n^2 = 1 - \frac{\omega_p^2}{\omega^2}$*
 - ❖ Lots of neutrals means high collision frequency (more energy absorption); the index of refraction is more complicated.
 - ❖ Absorption of MHz waves (next)
 - ❖ Reflection of waves below ~ 100 kHz; VLF waves (below ~ 50 kHz) used for long-range communications with **submarines**

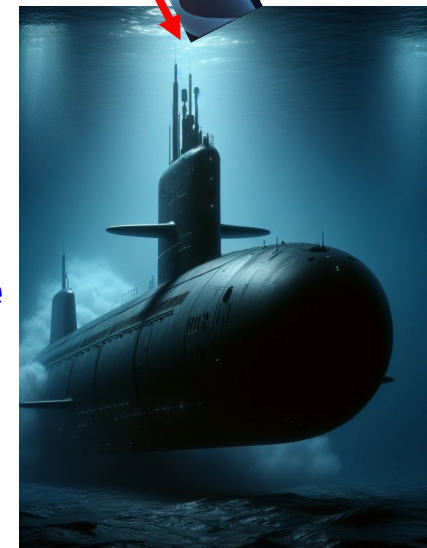


Earth-Based Very Low Frequency (VLF) Transmitters



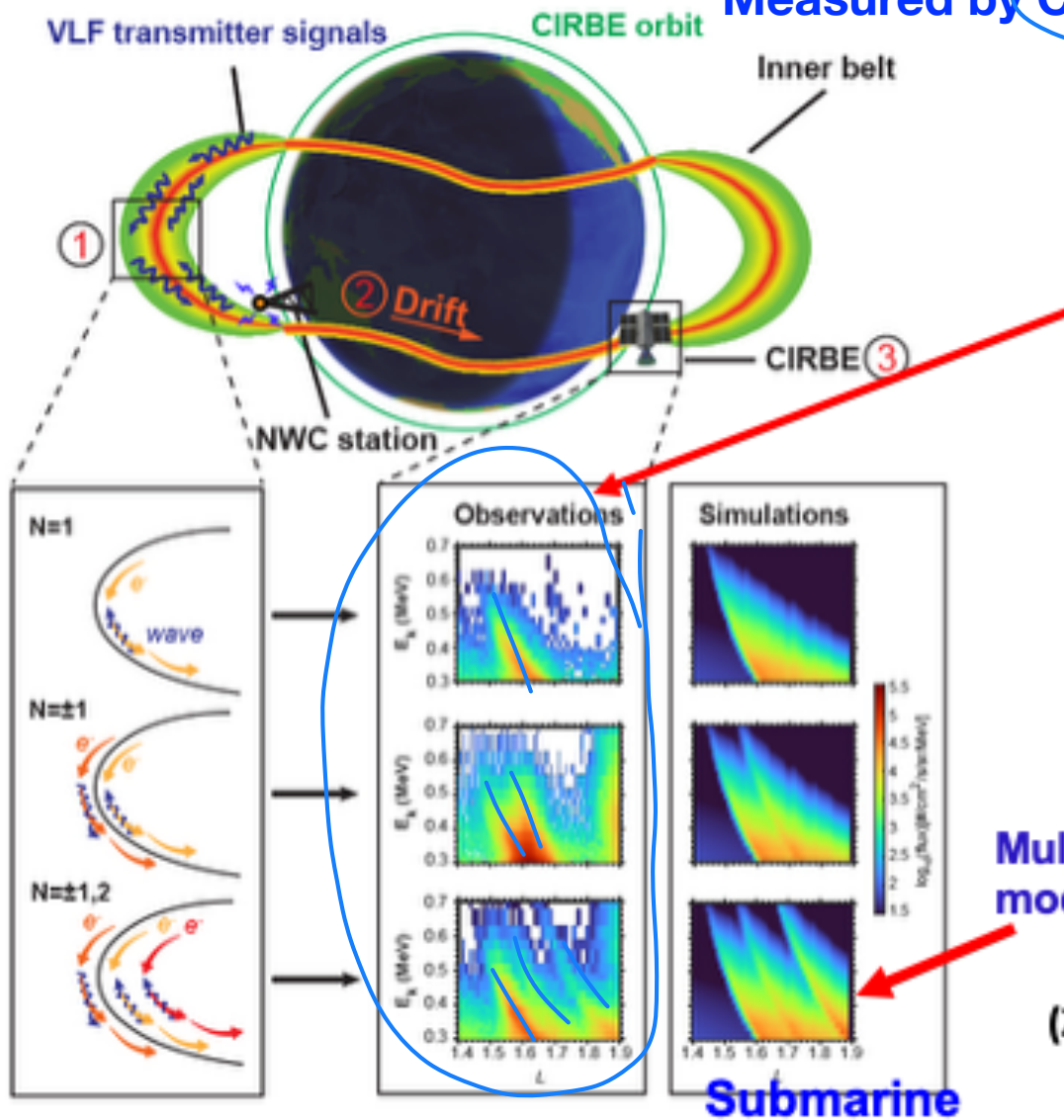
Power: 1 MW
Height: 387 meters
Frequency: 19.8 kHz

Submarine



Earth-Based VLF Transmitters Trigger Precipitation of Inner Radiation Belt Electrons

Measured by CIRBE/REPTile-2



Multiple wisps observed and modeled for the first time

(Zheng Xiang, X. Li et al.,
AGU Advances, 2024)

Previous works focused on single wisp, e.g.,
Imhof et al., 1983; Sauvaud et al., 2008; Selesnick et al., 2013; Cunningham et al., 2022

*** high-energy resolution and clean measurements required to capture the multiple wisps ***

Waves in Plasmas

- ❖ We need three equations to describe wave propagation in a cold plasma.
Ampere's Law and Faraday's Law (i.e. Maxwell's equations):

$$\begin{aligned}\nabla \times \vec{B} &= \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{d\vec{E}}{dt} \quad \rightarrow \frac{d}{dt} \rightarrow i\omega \sim i\omega \quad \Rightarrow A e^{i(\omega t - kx)} \\ \nabla \times \vec{E} &= -\frac{d\vec{B}}{dt} \quad \rightarrow \vec{\nabla} \times \rightarrow i\vec{k} \times \quad k = \frac{2\pi}{\lambda}\end{aligned}$$

- ❖ And the Langevin equation (simply $\underline{F = ma}$):

$$m_e \frac{d\vec{v}_e}{dt} = \sum \vec{F} = q_e \vec{E} + q_e (\vec{v} \times \vec{B}) + \underbrace{\nu m_e \vec{v}_e + \dots}_{\text{collision freq.}}$$

$$\vec{J}_e = q_e n_e \vec{v}_e \quad \nu = \frac{k_c}{\omega}$$

Index of refraction in a cold plasma (in a B field)

$$n^2 = 1 - \frac{\omega_p^2}{\omega^2}$$

❖ In general, given by the Appleton-Hartree equation:

$$n^2 = 1 - \frac{X}{1 - iZ - \frac{\frac{1}{2}Y^2 \sin^2 \theta}{1 - X - iZ} \pm \frac{1}{1 - X - iZ} \left(\frac{1}{4}Y^4 \sin^4 \theta + Y^2 \cos^2 \theta (1 - X - iZ)^2 \right)^{1/2}}$$

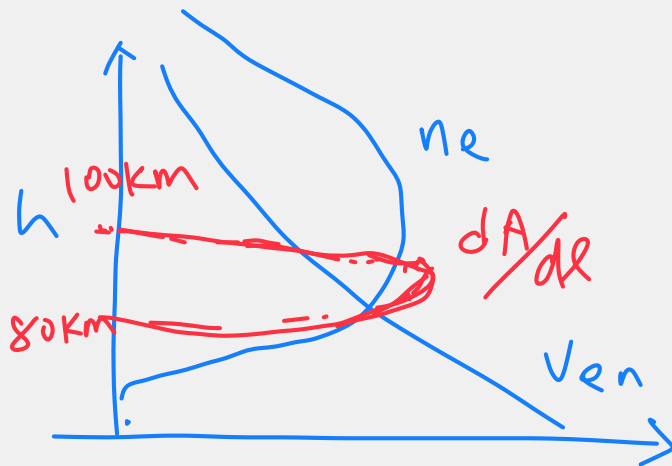
$X = \frac{\omega_p^2}{\omega^2}$ — plasma freq.
 $Y = \frac{\omega_c}{\omega}$ — gyrofrequency
 $Z = \frac{\nu}{\omega}$ — collision freq.

angle between $\vec{B}_0$ & $\vec{k}$

make n complex
 $n = \alpha + i\beta$

D-region absorption

- ❖ As electrons get excited by waves with $f < f_c$, they collide with neutrals
- ❖ Some of the EM wave energy gets transferred to heat; radio waves suffers stronger absorption
- ❖ How much absorption?



$$\frac{dA}{dh} \left(\frac{dB}{m} \right) \sim 4.6 \times 10^{-5}$$

$$\frac{n_e v_{en}}{\omega^2}$$

$$\frac{n_e v_{en}}{v_{en}^2 + (\omega \pm \omega_c)^2} \approx \frac{2B}{m_e}$$

$\omega > \omega_c$
 $> v_{en}^2$

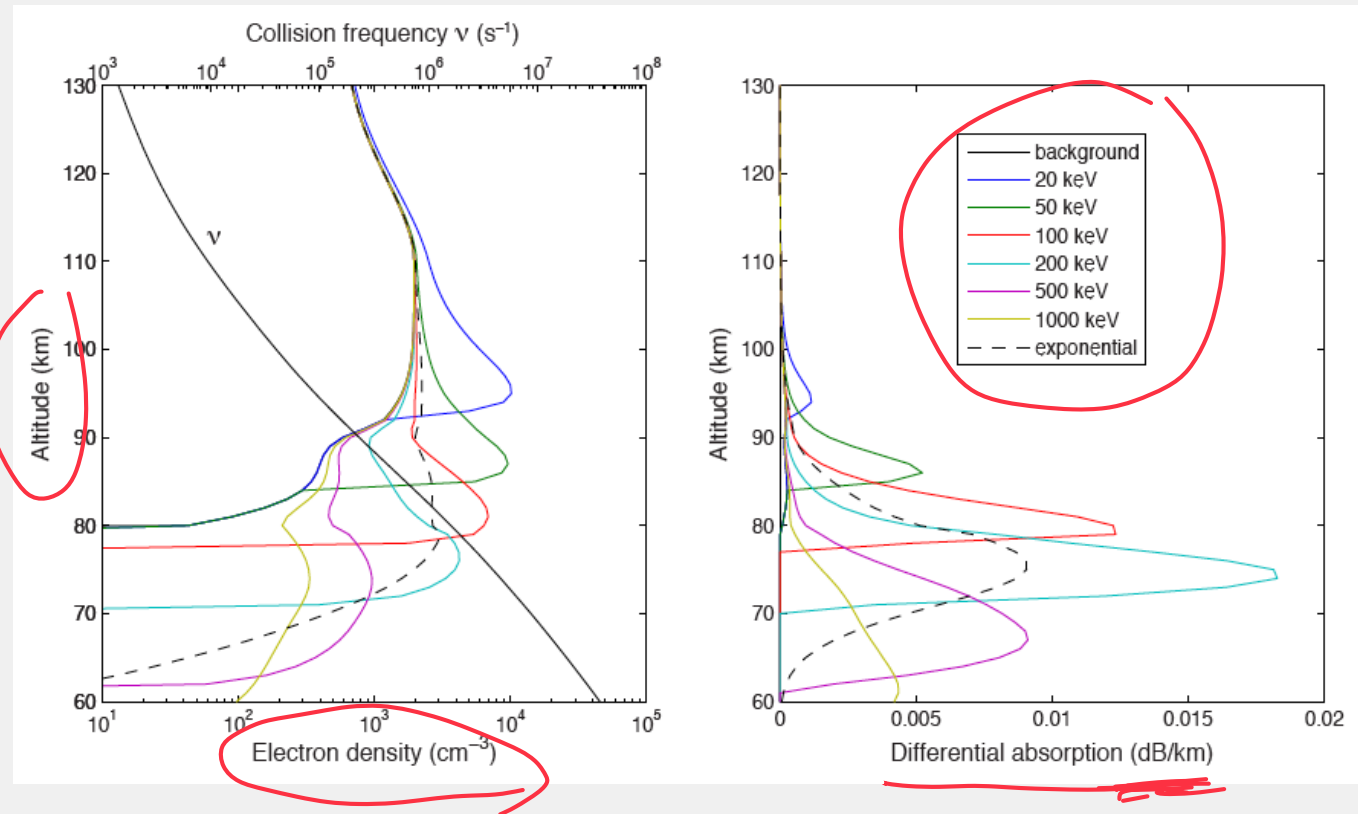
$v_{en} \propto n_i$
neutral density

Collision Frequency?

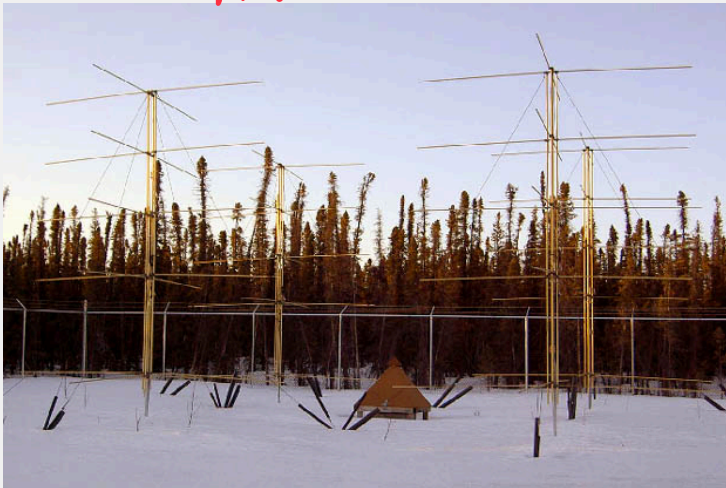
- ❖ Electrons (few) randomly collide with neutrals (many)
 - ❖ Radio wave energy converts to electron kinetic energy and then to neutral thermal energy (i.e., neutrals are "heated")
 - ❖ This is collisional heating, and a sink for radio wave energy
-
- ❖ Collision frequency depends on neutral density (N₂, O₂) and on electron temperature

D-region absorption

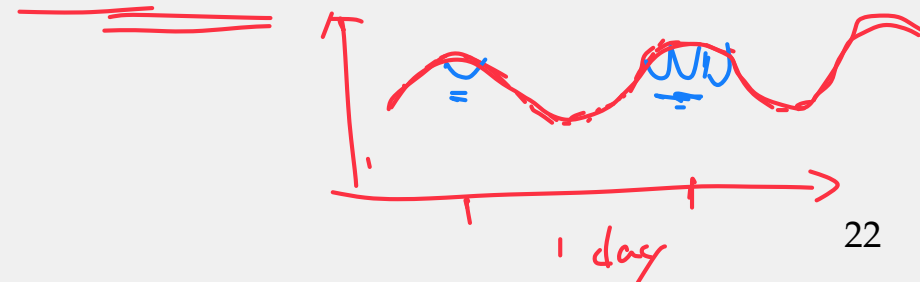
- ❖ Shown here:
absorption of 30
MHz radio wave due
to electron
precipitation from
the radiation belts



Alaska

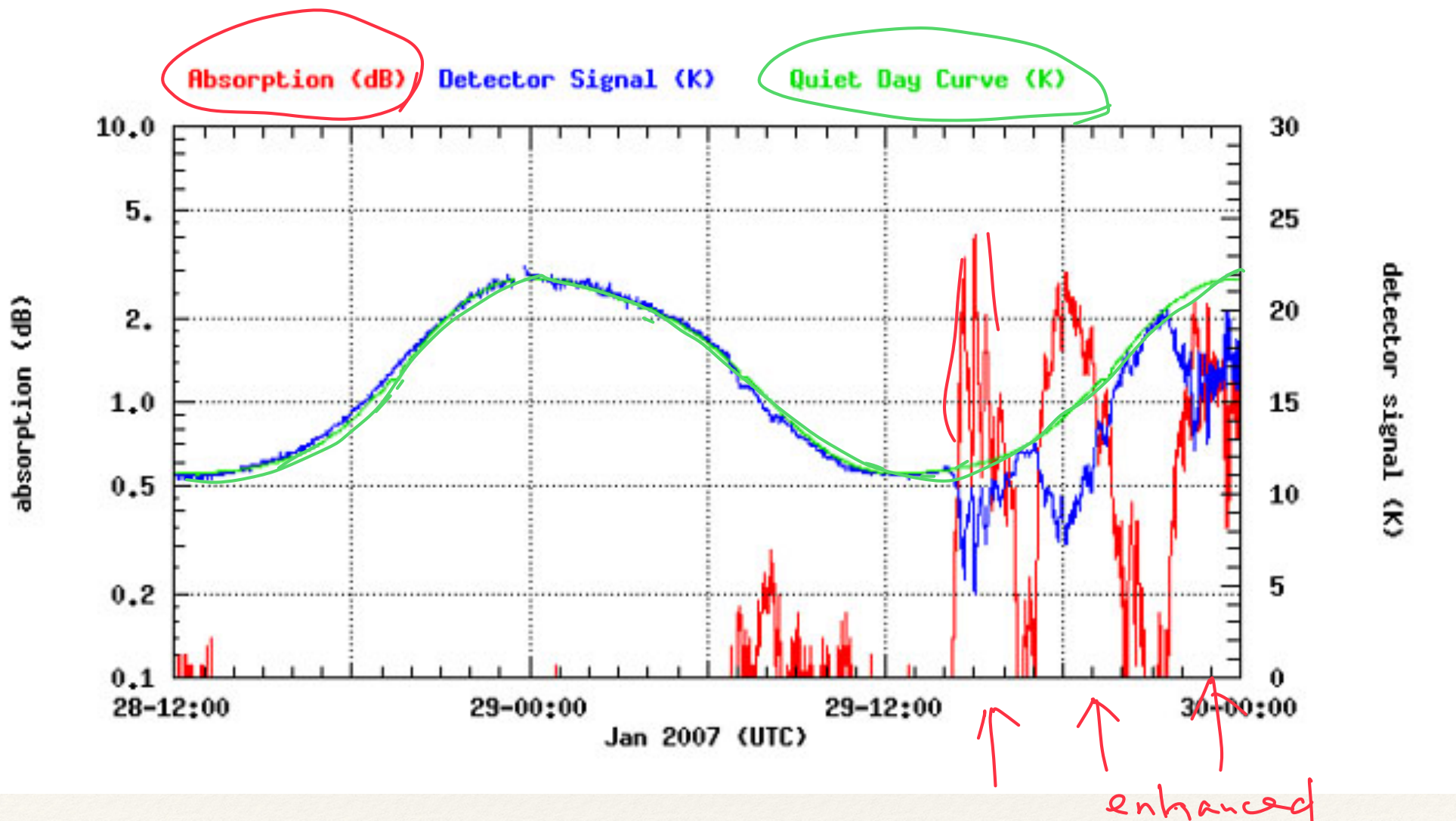


- ❖ **Riometer (Relative Ionospheric Opacity Meter)**: passive instrument that measures D-region absorption by monitoring cosmic noise at ~ 30 MHz



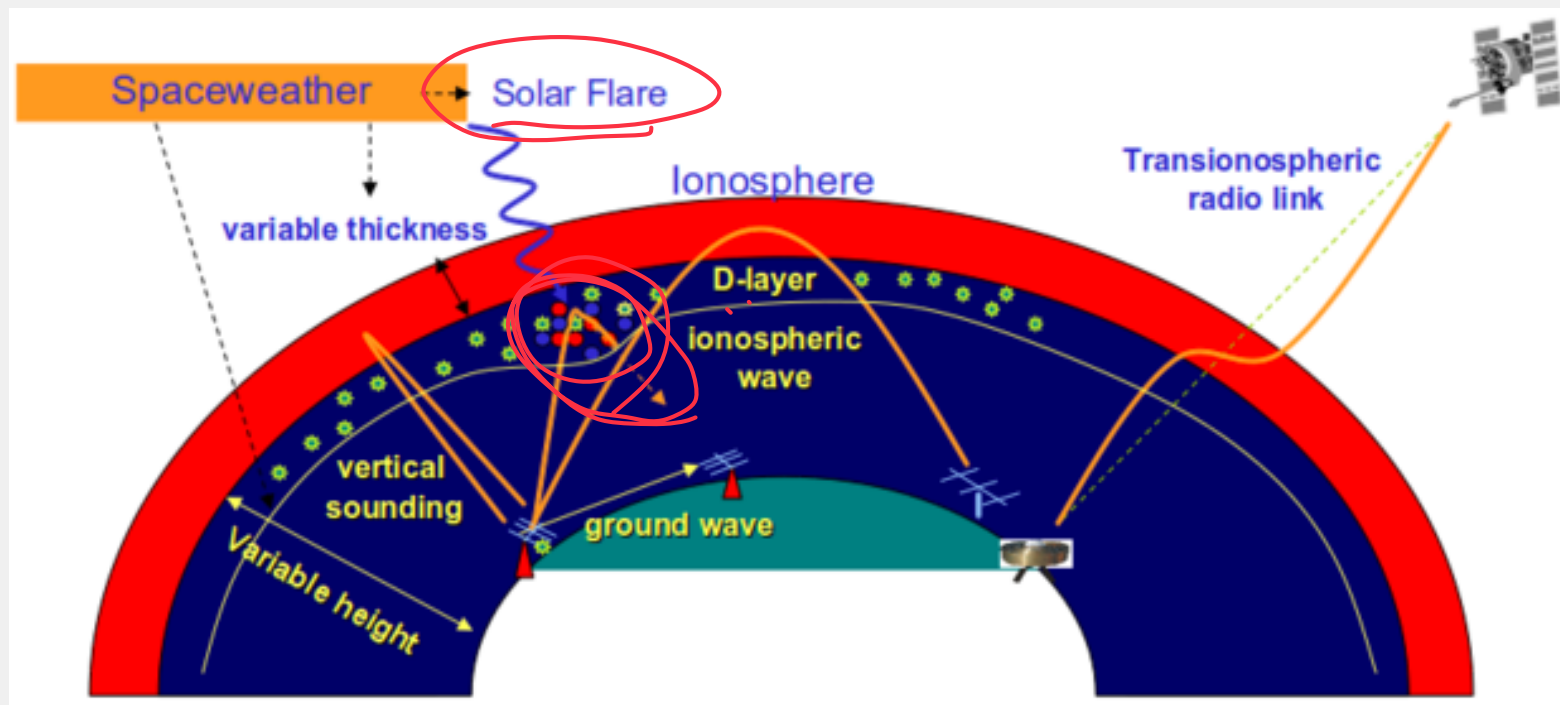
- ❖ An example of **Riometer** measurements

(HARP)



Sudden Ionospheric Disturbances (SIDs)

- ❖ SID is ionospheric response to a solar flare (EUV and X-rays)
- ❖ EUV and X-rays ionize the D-region, causing a huge increase in D-region electron density (orders of magnitude)
- ❖ Higher n_e leads to higher radio wave absorption
- ❖ Lower D-region reflection height perturbs VLF signals



Short-Wave Fade

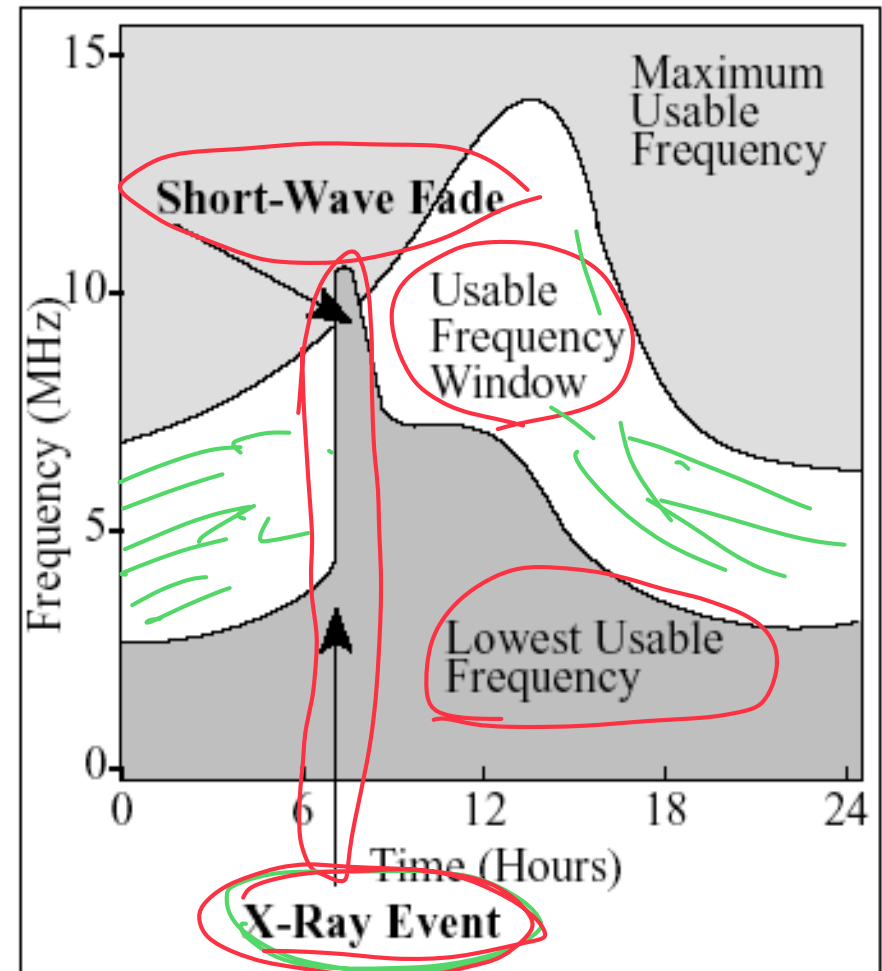
❖ Issue for over-the-horizon radar

- ❖ There is a maximum frequency we can use, above which waves pass through the F-region

- ❖ There is a minimum frequency we can use, because

$$\frac{dA}{dl} = 4.6 \times 10^{-5} \frac{n_e \nu}{\omega^2}$$

- ❖ lower frequencies suffer too much absorption
- ❖ Usable "Frequency Window"
- ❖ After a major X-ray flare, Absorption can increase to prevent any useful communication (over-the-horizon)



GPS and TEC

$$f \sim 1 - 1.54 \text{ Hz}$$

- ❖ Even for frequencies above f_c , the ionosphere introduces some interesting effects

- ❖ Small change in index of refraction from $n^2 = 1 - \frac{\omega_p^2}{\omega^2}$

- ❖ Change in '**path length**': $\Delta l_{\text{iono}} = - \frac{40.3}{f^2} \text{TEC}$

- ❖ Where TEC is total electron content, integrated along signal path

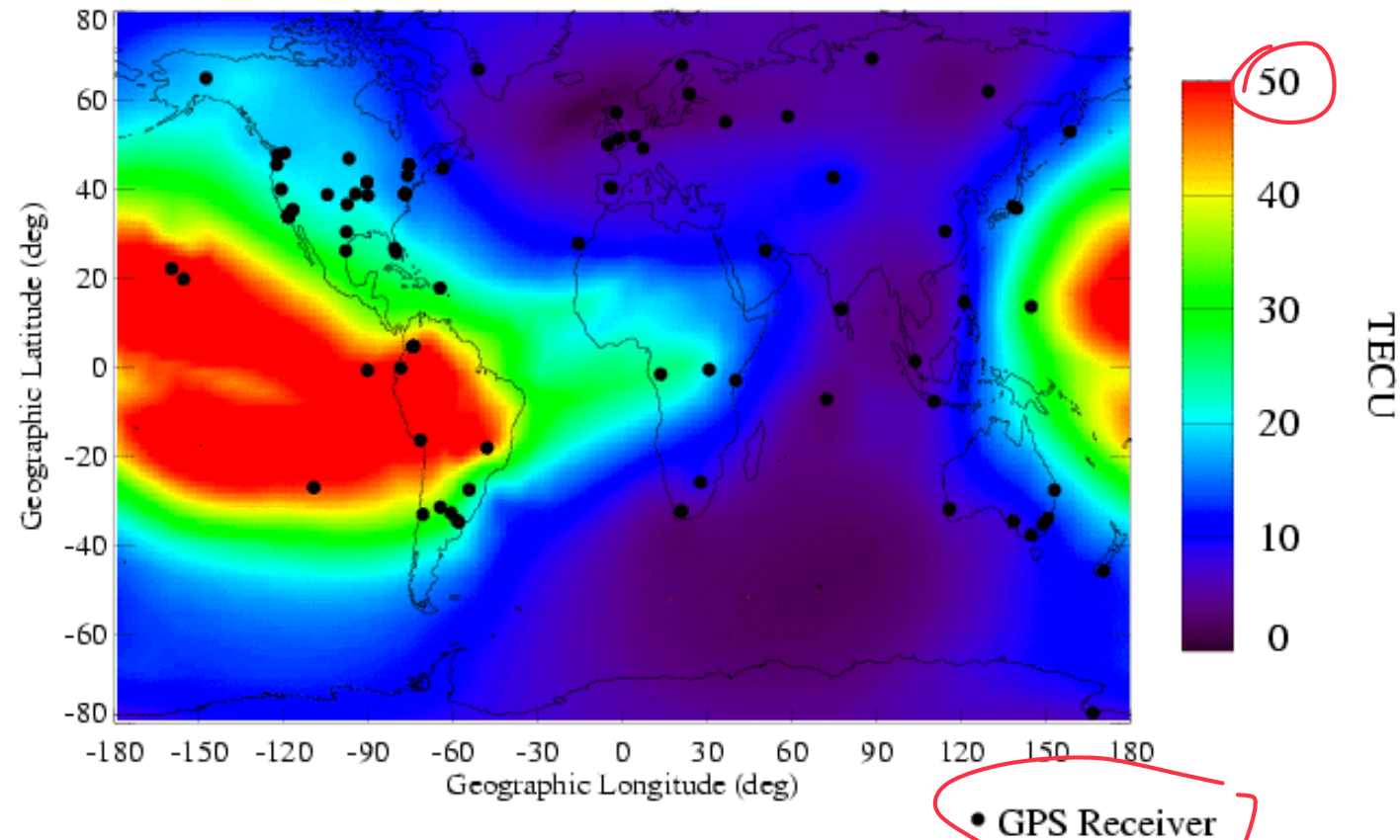
$$\text{TEC} = \int_0^\infty n_e(h) dh \quad (\text{m}^{-2})$$

TEC maps

- ❖ 1 TECU = 10^{16} el/m²
- ❖ receivers all over the Earth's surface; 20+ satellites to provide pierce-points
- ❖ Interpolate results onto 2D (or 3D) map

02/09/05
22:40 UT

Ionospheric TEC Map



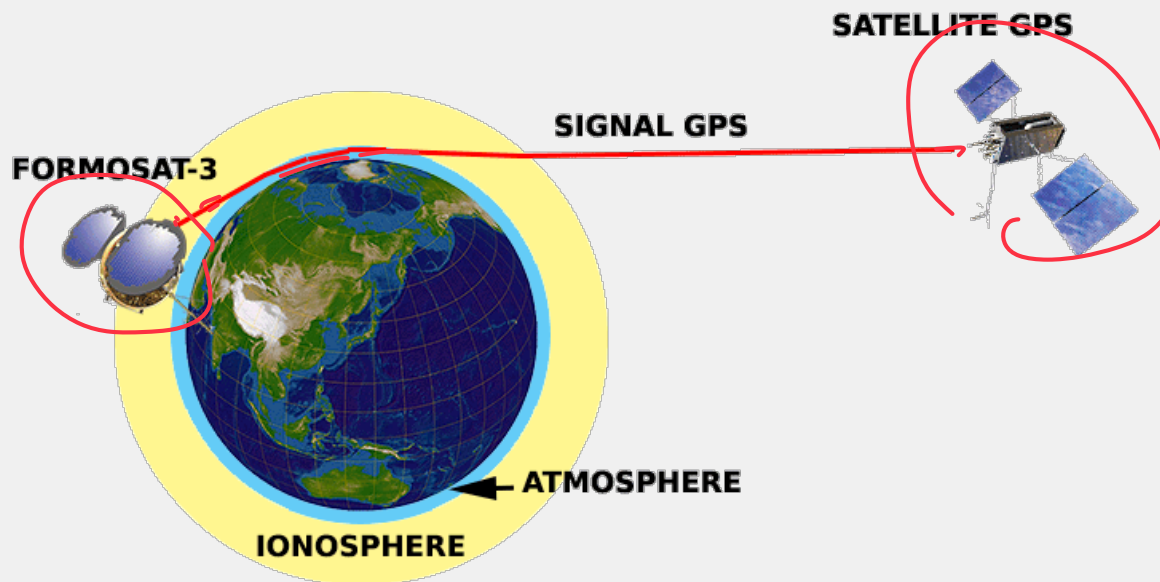
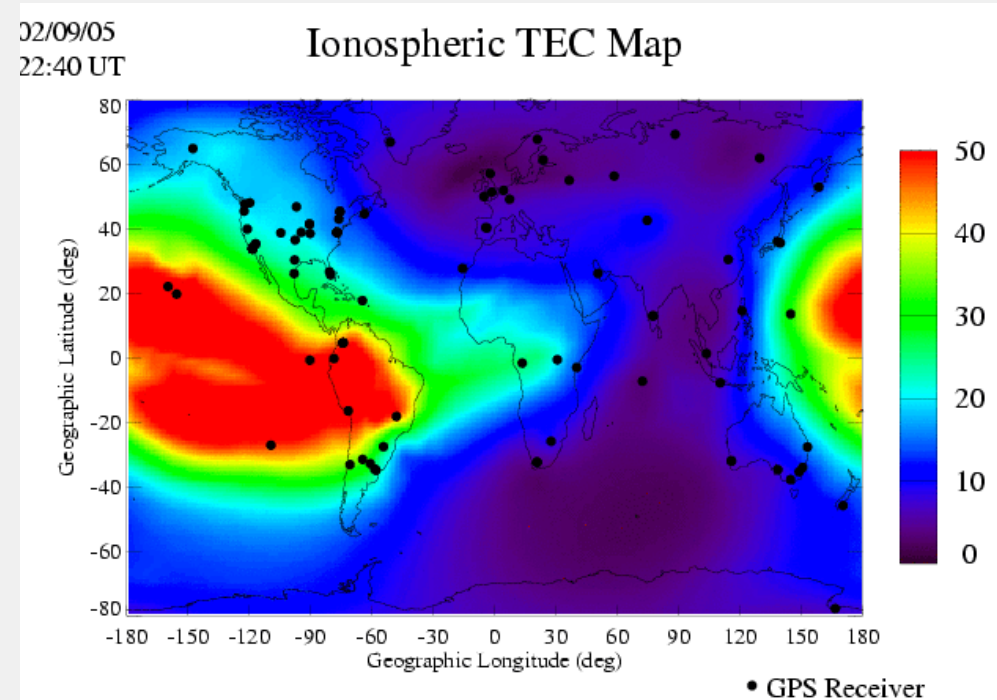


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Radio Occultation

GPS and COSMIC

- ❖ Use radio occultation (path delay) to measure TEC
- ❖ Take a ton of TEC measurements and throw them into a giant inversion model to get 4D ionosphere
- ❖ Also use spacecraft-to-spacecraft measurements: COSMIC



- ❖ COSMIC is GPS receiver instrument on Taiwanese FORMOSAT-3 spacecraft
- ❖ Takes 1500 radio occultation measurements per day



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The Ionosphere: Missions and Movies
